

Monetary Policy According to HANK

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Monetary transmission in NK models

Representative Agent model (RANK) I

Model structure: CRRA utility from C_t with parameter γ , discount rate ρ , representative firm produces output using labor $Y = N$.

$$\frac{\dot{C}_t}{C_t} = \frac{1}{\gamma}(r_t - \rho) \quad (\text{Euler equation})$$

$$r_t = \rho + e^{-\eta t}(r_0 - \rho) \quad (\text{analytic case})$$

Later we will introduce government which receives (pays) lump-sum transfers (taxes) T_t and issues bonds B_t .

Optimal consumption function $\mathcal{C}(\{r_t, Y_t, T_t\}_{t \geq 0})$

Representative Agent model (RANK) II

Consumption response to interest rate change:

$$dC_0 = \underbrace{\int_0^\infty \frac{\partial C_0}{\partial r_t} dr_t dt}_{\text{direct response to } r} + \underbrace{\int_0^\infty \frac{\partial C_0}{\partial Y_t} dY_t dt}_{\text{indirect effect due to } Y}$$

$$\frac{d \log C_0}{dr_0} = \frac{-1}{\gamma \eta} = \frac{-1}{\gamma \eta} \left(\underbrace{\frac{\eta}{\rho + \eta}}_{\text{direct response to } r} + \underbrace{\frac{\rho}{\rho + \eta}}_{\text{indirect effect due to } Y} \right)$$

With government:

$$dC_0 = \underbrace{\int_0^\infty \frac{\partial C_0}{\partial r_t} dr_t dt}_{\text{direct response to } r} + \underbrace{\int_0^\infty \left(\frac{\partial C_0}{\partial Y_t} dY_t + \frac{\partial C_0}{\partial T_t} dT_t \right) dt}_{\text{indirect effect}}$$

$$\frac{d \log C_0}{dr_0} = \frac{-1}{\gamma \eta} = \frac{-1}{\gamma \eta} \left(\underbrace{\frac{\eta}{\rho + \eta} \left(1 - \rho \gamma \frac{B_0}{\bar{Y}} \right)}_{\text{direct response to } r} + \underbrace{\frac{\rho}{\rho + \eta} + \frac{\eta}{\rho + \eta} \rho \gamma \frac{B_0}{\bar{Y}}}_{\text{indirect effect}} \right)$$

Non-Ricardian Hand-to-Mouth (TANK) model

Introducing **fraction Λ** of **constrained households** without access to financial markets often called **rule-of-thumb**.

$$\frac{d \log C_0}{dr_0} = \frac{-1}{\gamma\eta} = \frac{-1}{\gamma\eta} \left(\underbrace{(1-\Lambda) \frac{\eta}{\rho + \eta}}_{\text{direct response to } r} + \underbrace{(1-\Lambda) \frac{\rho}{\rho + \eta} + \Lambda}_{\text{indirect effect due to } Y} \right)$$

fiscal response with some transfers to hand-to-mouth $\rightarrow$ bigger reaction to monetary policy, Ricardian equivalency does not hold

Direct effects (%)	$B = 0$	$B > 0$	S-W	$B, K > 0$
RANK	99	98	99	94
TANK	69	57	—	50

HANK model

Household I

HANK model with two assets **liquid** b_t and **illiquid** a_t .

Households face **idiosyncratic** Markov process for **productivity** z_t , die with Poisson intensity ζ , replaced with asset-free household.

$$\mathbb{E}_0 \int_0^\infty e^{-(\rho+\zeta)t} u(c_t, \ell_t) dt$$

Assets evolve according to:

$$\dot{b}_t = (1 - \tau_t) w_t z_t \ell_t + r_t^b(b_t) b_t + T_t - d_t - \chi(d_t, a_t) - c_t,$$

$$\dot{a}_t = r_t^a a_t + d_t$$

$$b_t \geq -\underline{b}, \quad a_t \geq 0.$$

Illiquid wealth composition

Illiquid savings a_t are composed out of: capital k_t and equity shares s_t :

$$a_t = k_t + q_t s_t$$

Dynamics given by:

$$\dot{k}_t + q_t \dot{s}_t = (r_t^k - \delta)k_t + \Pi_t s_t + d_t$$

Return of combined asset:

$$\frac{\Pi_t + \dot{q}_t}{q_t} = r_t^k - \delta \equiv r_t^a \quad (\text{no arbitrage})$$

Closing model

Everything else as in standard NK model with capital and price adjustments costs (Rotemberg 1982).

$$\left(r_t^a - \frac{\dot{Y}_t}{Y_t}\right) \pi_t = \frac{\epsilon}{\theta} \left(mc_t - \frac{\epsilon - 1}{\epsilon}\right) + \dot{\pi}_t \quad (\text{NK Phillips curve})$$

Monetary authority:

$$i_t = \bar{r}^b + \phi \pi_t + \epsilon_t$$

Government's IGBC:

$$\dot{B}_t + G_t + T_t = \tau_t \int w_t z \ell_t(a, b, z) d\mu_t + r_t^b B_t$$

Important equilibrium object: measure of agents $\mu_t(a, b, z)$ - high dimensionality, g.e.:

$$B_t + \int b(a, b, z) d\mu_t(a, b, z) = 0 \quad (\text{illiquid asset market clearing})$$

$$N_t = \int z \ell_t(a, b, z) d\mu_t \quad (\text{labor market clearing})$$

Monetary Transmission in HANK

$$\begin{aligned} dC_0 = & \underbrace{\int_0^\infty \frac{\partial C_0}{\partial r_t^b} dr_t^b dt}_{\text{direct effect}} \\ + & \underbrace{\int_0^\infty \left(\frac{\partial C_0}{\partial w_t} dw_t + \frac{\partial C_0}{\partial r_t^a} dr_t^a + \frac{\partial C_0}{\partial \tau_t} d\tau_t + \frac{\partial C_0}{\partial T_t} dT_t \right) dt}_{\text{indirect effect}} \end{aligned}$$

Calibration

Calibration I

Liquid (B^h)		Illiquid ($A = K + q$)		Total
Revolving consumer debt	-0.03	Net housing	1.09	
Deposits	0.23	Net durables	0.22	
Corporate bonds	0.04	Corporate equity	1.02	
Government bonds	0.02	Private equity	0.59	
Total	0.26		2.92	3.18

Continuous-time earning dynamics:

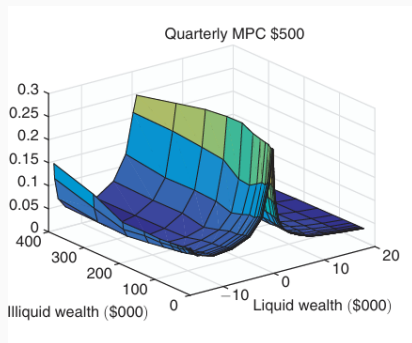
$$\log z_{it} = z_{1,it} + z_{2,it}$$

with Poisson rates λ_1, λ_2 . Interpretation: periodic temporary shock, large persistent "career" shock.

Calibration II

Description		Value	Target/source
<i>Preferences</i>			
ζ	Death rate	1/180	Avg. lifespan 45 years
$1/\gamma$	Intertemporal elasticity of subst.	1	
$1/\nu$	Frisch elasticity of labor supply	1	
φ	Disutility of labor	2.2	Avg. hours worked equal to 1/2
ρ	Discount rate (p.a.)	5.1%	Internally calibrated
<i>Production</i>			
ϵ	Demand elasticity	10	Profit share of 10 percent
θ	Price adjustment cost	100	Slope of Phillips curve, $\epsilon/\theta = 0.1$
α	Capital share	0.33	
δ	Steady-state depreciation rate (p.a.)	7%	
<i>Government</i>			
τ	Proportional labor tax	0.30	
T	Lump-sum transfer (rel. GDP)	0.06	40% hh with net govt. transfer
<i>Monetary Policy</i>			
ϕ	Taylor rule coefficient	1.25	
$\bar{r}^b$	Steady-state real liquid return (p.a.)	2%	
<i>Unsecured borrowing</i>			
r^{borr}	Borrowing rate (p.a.)	8%	Internally calibrated
$\underline{b}$	Borrowing limit	\$16,500	1× quarterly labor income
<i>Adjustment cost function</i>			
χ_0	Linear component	0.0438	Internally calibrated
χ_1	Convex component	0.956	Internally calibrated
χ_2	Convex component	1.402	Internally calibrated
$\underline{a}$	Min a in denominator	\$1,000	

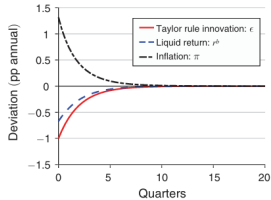
Calibration III



	Data	Model	Moment	Liquid wealth		Illiquid wealth	
				Data	Model	Data	Model
Mean illiquid assets	2.92	2.92	Top 0.1 percent share	17	2.3	12	7
Mean liquid assets	0.26	0.23	Top 1 percent share	47	18	33	40
Frac. with $b = 0$ and $a = 0$	0.10	0.10	Top 10 percent share	86	75	70	88
Frac. with $b = 0$ and $a > 0$	0.20	0.19	Bottom 50 percent share	-4	-3	3	0.1
Frac. with $b < 0$	0.15	0.15	Bottom 25 percent share	-5	-3	0	0
			Gini coefficient	0.98	0.86	0.81	0.82

Results I

Panel A. Monetary shock, interest rate, inflation



Panel B. Aggregate quantities

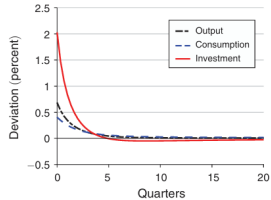
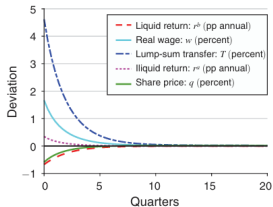
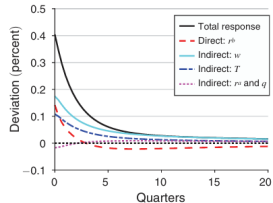


FIGURE 3. IMPULSE RESPONSES TO A MONETARY POLICY SHOCK
(A Surprise, Mean-Reverting Innovation to the Taylor Rule)

Panel A. Prices



Panel B. Consumption decomposition

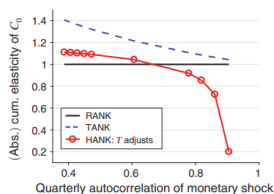


Results II

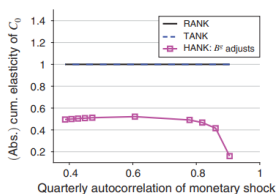
TABLE 8—IMPORTANCE OF FISCAL RESPONSE TO MONETARY SHOCK

	<i>T</i> adjusts (1)	<i>G</i> adjusts (2)	τ adjusts (3)	B^s adjusts (4)
Change in r^b (pp)	−0.28	−0.23	−0.33	−0.34
Elasticity of Y	−3.96	−7.74	−3.55	−2.17
Elasticity of I	−9.43	−14.44	−8.80	−5.07
Elasticity of C	−2.93	−2.80	−2.75	−1.68
Partial eq. elasticity of C	−0.55	−0.60	−0.56	−0.71
<i>Component of percent change in C due to</i>				
Direct effect: r^b	19	21	20	42
Indirect effect: w	51	81	62	49
Indirect effect: T	32	—	—	9
Indirect effect: τ	—	—	18	—
Indirect effect: r^a and q	−2	−2	0	0

Panel A. *T* adjusts



Panel B. B^s adjusts



The Household Problem (HJB equation):

- PDE $\rightarrow$ finite difference,
- discretize state space,
- upwind scheme
- discretized operator (matrix) is sparse and tridiagonal $\rightarrow$ easy to solve

Evolution of distribution (KFE equation):

- adjoint to HJB